

Design Pitfalls in Interactive Sales Robots: A Systematic Analysis and Framework

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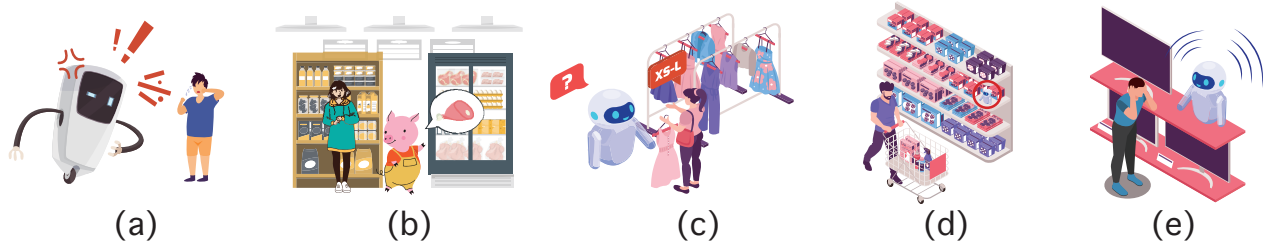


Figure 1: Five categories of design pitfalls. (a) Ethical and social violations, (b) Selection of robots unsuitable for the store or products, (c) Lack of robot functionality, (d) Environments that reduce opportunities for interaction, and (e) Behaviors that discourage continued interaction.

Abstract

This study proposes a systematic analysis of design pitfalls in interactive sales robots, derived from workshops and online surveys. We collected 423 ineffective behaviors of interactive sales robots through workshops and classified them into five major categories and 26 subcategories. These findings provide actionable insights for researchers and practitioners, emphasizing the importance of user-centered design to improve robot usability and adoption.

CCS Concepts

• **Human-centered computing** → **Interaction design theory, concepts and paradigms.**

Keywords

Human-Robot Interaction, Interactive Sales Robot, Design Pitfalls

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1 Introduction

Studies on human–robot interaction (HRI) have explored conversational social robots in fields such as elderly care [3] and education [2], with customer service being a key application. In customer service, conversational social robots are expected by both stores and customers to perform tasks such as product introductions and sales promotion [21]. These capabilities aim to address labor shortages and enhance the efficiency of store operations. This study defines robots performing these tasks as “interactive sales robots.” Studies on interactive sales robots have been conducted in various retail settings, including shopping malls [37], convenience stores [14], department stores [42], and bakeries [26].

Controlled experiments of interactive sales robots, in general, exclude negative factors, such as insulting customer interactions, to ensure valid evaluations, because such factors can lead to user disappointment and distrust, ultimately undermining long-term trust in robotic technologies [30]. Previous studies focused primarily on demonstrating the effectiveness of interactive sales robots, with insufficient discussion on design failures or ineffective behavioral patterns. This lack of focus is partly because researchers have prioritized the success of their experiments, treating these ineffective factors as tacit knowledge.

These ineffective patterns occasionally referred to as anti-patterns, have been studied and systematized in various fields including software development [16] and project management [19]. In addition, these systematized anti-patterns have been used to develop system bug prediction models [38]. Systematizing such ineffective patterns is a critical guideline for future system design. Although several of these ineffective design patterns are commonly acknowledged, creating a comprehensive list is essential to avoid design errors and

provide a foundation for widespread adoption. In contrast, although previous studies on interactive sales robots have focused on only their effectiveness, systematic discussions about design failures or ineffective behavioral patterns remain insufficient.

Therefore, this study aims to systematize these ineffective design patterns and provide guidelines for improving robot design. We define ‘*design pitfalls*’ as ineffective patterns specific to interactive sales robots that hinder their proper execution of intended roles. To systematize the design pitfalls, five workshops were conducted to gather ideas from users about undesirable robot behaviors. Based on the collected data, the design pitfalls were classified into five major categories, as shown in Figure 1.

This study makes a significant contribution by systematically categorizing design pitfalls in interactive sales robots. These falls are classified into five major categories and 26 subcategories, providing a comprehensive framework to help designers identify and address potential pitfalls. This framework serves as a foundational reference for improving the usability and effectiveness of interactive sales robots in real-world applications.

2 Exploratory Workshop

In this study, exploratory online workshops and an online survey were conducted to collect and systematize the design pitfalls of interactive sales robots. This section provides a detailed explanation of the workshop’s design, procedures, and methods for data collection and analysis.

2.1 Workshop Design and Objectives

The workshops were designed to identify potential design pitfalls in interactive sales robots. Participants were asked to consider “undesirable robot behaviors” in various scenarios and document these behaviors. The undesirable robot behaviors refer to actions by robots that users perceive as ineffective, inappropriate, or counter-productive in specific scenarios. These behaviors were later categorized as design pitfalls to provide actionable insights for improving robot design. This study focused on collecting opinions from actual users rather than from developers or corporate stakeholders [21], to extract design pitfalls from the perspective of the end-user by prioritizing user-centered design [22].

The participants comprised primarily students and staff from multiple university laboratories involved in robotics research. By targeting participants who are familiar with robotics, the workshop was expected to yield realistic and diverse ideas for the behavior and design of robots. A total of 13 participants were recruited and divided into five groups, each comprising 2–3 members. All participants were between the ages of 20 and 39. The workshop lasted approximately 90 min, and participants were compensated 1,500 JPY upon completion.

2.2 Workshop Procedure

The workshop was convened as per previous studies [39] and followed the procedure outlined below. The 90-min session began with a 5-min icebreaker activity to create an environment conducive to idea generation. Participants explored random scenarios combining robots, users, and environments to identify ‘undesirable robot behaviors’ for 5 min during one round, documented on Google



Figure 2: Example of how participants organized their ideas using Google Jamboard in each round of the workshop, with different sticky note colors for each experimental participant.

Jamboard. After each round, all participants shared their ideas, and additional perspectives were added if new insights emerged during discussions. This round process was repeated four times to collect various ideas under different scenarios.

Random scenarios combined the following Robots, Users, and Environments (e.g., Pepper interacting with a single adult in a restaurant).

Robots Aibo, Pepper, virtual agent (humanoid), and virtual agent (animal-like)

Users Single adult, two adults, single child, and parent–child pair

Environments Restaurant, supermarket, electronics store, car dealership, clothing store, general store, game shop, and travel agency

In cases involving two adults, participants were instructed to imagine themselves with a friend, partner, or family member. For parent–child pairs, participants considered scenarios from the perspective of the parent. Figure 2 illustrates an example of the use of Google Jamboard during the workshop to document the ideas of participants. Google Jamboard was selected owing to its ease of use for collaborative, real-time idea sharing in an online workshop environment.

2.3 Data Collection and Analysis

During the workshop, the collected ideas were recorded as unique and specific to the scenarios. These ideas were analyzed using the affinity diagram method [5], where similar ideas were grouped and labeled iteratively to extract themes and patterns, as the major categories and subcategories of design pitfalls. This categorization process was performed collaboratively by all authors. The joint effort ensured consistency in the interpretation of ideas while enhancing the transparency of the classification process.

To ensure that the final classification was comprehensive, GPT-4 [28] was employed as a supplementary tool. Based on the classification results created by the authors, GPT-4 was used to propose additional perspectives. The ideas generated by GPT-4 were meticulously reviewed and evaluated by the authors, and only those deemed original and practical were adopted. Although the suggestions from GPT-4 were referenced, GPT-4 was not directly involved in the classification process, maintaining the rigor of manual analysis.

To quantify the degree to which each subcategory represents design pitfalls, an online survey was conducted following the classification process. The survey aimed to numerically evaluate the extent to which each pitfall was necessary to avoid from the perspective of the customer. Respondents rated each subcategory on a

5-point Likert scale (1: Not bothersome, 5: Extremely bothersome). Subcategories were clarified with examples for context. The survey was administered via Google Forms, allowing participants to respond conveniently online. To enhance data validity, dummy questions were included to identify inattentive responses, and any submissions with inconsistencies in these control items were excluded from the final analysis. This approach ensured that the survey results reflected thoughtful and reliable evaluations of the identified behaviors.

3 Findings

A total of 423 ideas were collected and organized into five main categories from five workshops, along with 25 subcategories of design pitfalls. These pitfalls highlight common challenges and provide a framework for addressing issues in interactive robot design. In addition, GPT-4 was employed to generate 30 additional ideas, which were subsequently compared with the ideas collected from the workshops to identify overlaps or gaps. Consequently, one additional subcategory was added, leading to a final classification of five categories, 26 subcategories, and 60 specific examples. In this process, GPT-4 served a supportive role, with all final decisions made solely by the researchers. Five major categories were identified: (1) Ethical and social violations, (2) Robot selection unsuitable for stores or products, (3) Lack of robot functionality, (4) Environments that reduce interaction opportunities, and (5) Behaviors that discourage continued interaction. These results are summarized in Tables 1–5.

Furthermore, previous studies related to each subcategory were surveyed, and the findings are also included in Tables 1–5. The main focus or discussions of these previous studies include both direct and indirect correspondences to the specific examples listed in the subcategories. For instance, the previous study [1] aimed to convey the intentions or emotions of a robot though its gaze is indirectly aligned with the subcategory “*Untrustworthy or awkward gaze and facial expressions*” in Category 5 and is included as relevant related work.

In addition to the workshop results, an online survey was conducted to quantify the level of avoidance associated with each subcategory. The survey collected 211 responses, of which 98 were deemed valid after excluding inattentive submissions identified through dummy questions. Among the valid responses, the demographics of the participants included 58 males, 34 females, and one who preferred not to disclose their gender. The age distribution ranged from participants in their 20s (three individuals) to those aged 60 and above (seven individuals), with the largest representation from individuals in their 30s (25 individuals) and 40s (44 individuals). The findings from the survey are summarized in Tables 1–5, providing average scores as the perceived severity score for each subcategory. These scores offer a quantitative perspective on the degree to which each pitfall is perceived as undesirable.

4 Discussion

This study aimed to identify and categorize ineffective design patterns, referred to as design pitfalls, in interactive sales robots. The findings revealed key insights into perceived severity scores and

novel categories of design pitfalls. This section discusses the implications of these findings, compares them with those of previous studies, and outlines its limitations and directions for future studies.

4.1 Perceived Severity and Design Implications

The findings highlight that Category 1 (Ethical and social violations) often ranks higher in perceived severity scores compared with other categories. This result underscores the critical importance of addressing ethical and social violations in robot design. These issues pose significant risks to user trust and acceptance of robotic technologies if left unaddressed.

In contrast, Category 4 (Environments that reduce interaction opportunities) and Category 5 (Behaviors that discourage continued interaction) highlight less obvious design challenges. For instance, “*High pressure*” behaviors, such as overwhelming customers with excessive assistance or multiple robots [35], may cause discomfort and disengagement owing to insensitivity to user context. Similarly, “*Incoordination with human staff*” can result in robots appearing disconnected or unhelpful [26], potentially diminishing both customer trust and sales effectiveness.

4.2 Novel Design Pitfalls and Gaps in Previous Studies

This study highlights three novel design pitfalls: Aggressive behavior, unhygienic behavior, and unclear purpose. **Aggressive behavior:** Although ethical constraints limit empirical studies, addressing these behaviors in theoretical discussions is essential. Simulated environments could allow future studies to explore their impact without ethical concerns. **Unhygienic behavior:** Although previous studies emphasize positive design (e.g., costumes) [11], hygiene-related flaws, such as spreading dirt or material degradation, require further analysis. These issues harm trust and professionalism, particularly in contexts where cleanliness signals quality. **Unclear purpose:** This design pitfall highlights gaps in deploying robots for business. Customers may not understand the role of the robot, whether to inform, entertain, or sell, leading to disengagement. Clear signage or interaction prompts can address this.

4.3 Design Implications

This study provides practical insights for designing interactive sales robots by addressing design pitfalls in real-world applications. The results highlight the importance of incorporating user-centered design principles throughout the design process, and they also highlight the role of pitfalls in minimizing negative impacts, guiding early brainstorming, evaluating prototypes, and validating behavior. A synthesis of the five categories of design pitfalls reveals common underlying factors, including insufficient contextual awareness, limited adaptability to dynamic environments, and inadequate incorporation of user feedback in iterative design processes. To address these issues, future designs should prioritize adaptive learning mechanisms that enable robots to respond flexibly to diverse customer needs and situational changes. Additionally, fostering collaboration between robot designers and end-users can bridge the gap between technical feasibility and practical usability, ensuring that robots are not only functional but also contextually appropriate.

4.4 Limitations and Future Directions

Several limitations should be acknowledged. First, data collection primarily involved participants with robotics expertise, which may limit the generalizability of the findings to broader user groups. Future studies should recruit a more diverse sample, including participants representative of real retail environments, such as individuals from various age groups and backgrounds, to enhance generalizability.

Second, the identified design pitfalls were derived mainly from hypothetical scenarios and workshops rather than field studies. While this approach provides valuable insights, future research should validate these findings through real-world deployments of interactive robots in diverse commercial settings. Collecting observational data on how typical customers interact with robots will be crucial for developing best-practice guidelines and concrete redesign strategies to mitigate these design pitfalls.

Finally, although this study focused on identifying design pitfalls, further research is needed to explore strategies for maximizing the effectiveness of robot behaviors. In particular, studies targeting Categories 3 (Lack of robot functionality), 4 (Environments that reduce interaction opportunities), and 5 (Behaviors that discourage continued interaction) could provide actionable insights into optimizing robot functionality and interaction design.

Table 1: Category 1: Ethical and social violations. This table lists subcategories with their perceived severity scores from the survey, highlighting scores greater than the overall mean. Relevant previous studies are cited to provide context for each subcategory.

Subcategory (Perceived severity score)	Examples
Deceptive design (e.g., [34]) (4.35)	•Deceiving and lying to customers
Inappropriate responses to children (e.g., [9]) (4.28)	•Recommending overly expensive products to children •Introducing products harmful to the development of children
Aggressive behavior (4.80)	•Ignoring dangerous situations involving children or elderly •Engaging in harassment, discriminatory remarks, or rejecting customers •Exhibiting violent behaviors, such as threatening •Overreacting in ways that could harm people or products
Privacy infringement (e.g., [7]) (4.67)	•Collecting unnecessarily detailed personal information •Exhibiting privacy infringement functionality, such as accidentally entering fitting rooms, capturing inappropriate images (e.g., under skirts)
Unhygienic behavior (4.37)	•Appearing in dirty or tattered clothing •Engaging in actions that make the store or products dirty
Rule violations (e.g., [6]) (3.85)	•Violating policies established by the developer of the robot

Table 2: Category 2: Robot selection unsuitable for stores or products. This table lists subcategories with their perceived severity scores from the survey, highlighting scores greater than the overall mean. Relevant previous studies are cited to provide context for each subcategory.

Subcategory (Perceived severity score)	Examples
Unsuitable selection (e.g., [13]) (3.22)	•Evoking a sense of vivid realism leading to discomfort of the audience, such as a pig-shaped agent recommending pork products •Conflicting concepts of stores/products and robots, such as a robot with a cheap appearance recommending luxury products, a robot with an overly childish appearance introducing insurance products •Drawing attention away from the product owing to the size or behavior of the robot being overly prominent

Table 3: Category 3: Lack of robot functionality. This table lists subcategories with their perceived severity scores from the survey, highlighting scores greater than the overall mean. Relevant previous studies are cited to provide context for each subcategory.

Subcategory (Perceived severity score)	Examples
Unimplemented / Insufficient technical requirements (e.g., [20]) (3.52)	•Insufficient capabilities in image processing, speech processing, dialogue control, hardware, and networking (details omitted owing to the broad range of technical fields). •Incapacity to process reservations or payments while providing customer service or recommendations •Inadequate customer guidance owing to robots not being able to move or insufficient coordination between multiple robots •Requiring customers to assist the robot in tasks such as retrieving products from high shelves •Lacks multilingual support •Inappropriate amount of information, such as explaining cheap snacks in detail and not explaining expensive products well
Services falling below user expectations (e.g., [4]) (3.70)	•Failing to grasp store conditions, such as product locations, flash sale information, or sold-out products •Inability to compare products or provide knowledge about related items •Ignoring the interests or feedback of customers and providing recommendations disregarding the preferences of customers •Providing stereotypical recommendations based on appearance or gender alone •Lacking diversity in recommendations by relying solely on trends or surplus stock

Table 4: Category 4: Environments that reduce interaction opportunities. This table lists subcategories with their perceived severity scores from the survey, highlighting scores above the overall mean. Relevant previous studies are cited to provide context for each subcategory.

Subcategory (Perceived severity score)	Examples
Unclear purpose (2.88)	•Role of the robot is unclear, such as not clearly indicated by signs or other decorations
Inappropriate placement (e.g., [25]) (3.76)	•Environment where the robot embarrasses customers by making them stand out •Environment where interaction with one customer interferes with other customers
Hard to notice environment (e.g., [8]) (2.38)	•Placement where the robot is hard to notice, such as at the feet of customers •Robot is excessively small in size
Inaccessibility (e.g., [29]) (3.31)	•Lack of multimodal interaction design, such as voice dialogue only •Positioned at a height inaccessible to children •Unclear methods for interacting with the robot

Table 5: Category 5: Behaviors that discourage continued interaction. This table lists subcategories with their perceived severity scores from the survey, highlighting scores above the overall mean. Relevant previous studies are cited to provide context for each subcategory. Behaviors generated by GPT-4 are labeled with (GPT).

Subcategory (Perceived severity score)	Examples
Untrustworthy or awkward gaze and facial expressions (e.g., [1, 18]) (3.07)	•Lack of alignment with robot gaze and person/object it is observing •Mismatched facial expressions and lip-syncing, that is, verbal and non-verbal actions do not align •Unnatural expressions, such as lack of a smile and expressionless, or perpetually smiling
Inappropriate volume (e.g., [12]) (3.58)	•Unadjusted volume, such as excessively loud or quiet
Wrong conversation target (e.g., [40]) (3.64)	•Engaging with people uninterested in robots or ignoring those interested in robots •Spending excessive time with one customer, neglecting others
Incomprehension of customer situations (e.g., [10]) (3.50)	•Inability to estimate relationships within a group of customers •Failing to recognize previously purchased items or other contextual details such as window shopping
Lack of expressive variety (e.g., [15, 31]) (2.42)	•Limited patterns of motion •Failing to follow communication norms, such as greetings or responding to being touched or called “cute”
Discomforting manner of speaking (e.g., [17]) (3.20)	•Inappropriately formal or casual speech or speech mismatched with the mood of the customer •Conversation cannot proceed without being customer-led
Inappropriate dialogue content (e.g., [24]) (3.20)	•Using overly complex words not suited to the knowledge of the customer •Speaking about experiences robots cannot understand, e.g., “This food tastes delicious.”
Unnecessary functions in customer service (e.g., [33]) (3.55)	•Performing actions unrelated to customer service, such as dancing, sleeping, or asking to be petted •Engaging in disruptive actions, such as running around or touching customers
Inappropriate attitudes in customer service (e.g., [27]) (4.11)	•Exhibiting unwelcoming attitudes, such as being arrogant, unfriendly, or overly mechanical •Exhibiting overly exaggerated emotional expressions
High pressure (e.g., [36]) (4.28)	•High-pressure persuasion, such as with multiple robots and comparing customers •Providing excessive customer service during undesired moments, such as while fitting clothes
Inconsistent characterization (e.g., [41]) (3.01)	•Acting contrary to the character design of the robot, such as mismatched actions, speech, or voice •Engaging in speech overly aligned with its character, such as not speaking human language •Inconsistency in the personality or actions of the robot
Inappropriate speed (e.g., [32]) (3.03)	•Moving either excessively quickly or slowly
Incoordination with human staff (e.g., [26]) (3.90)	•Lack of human staff support even for selling high-value items •Exhibiting poor relationships with human staff
(GPT) Lack of growth (e.g., [23]) (3.59)	•Failing to leverage past mistakes or feedback

References

- [1] Henny Admoni and Brian Scassellati. 2017. Social Eye Gaze in Human-Robot Interaction: A Review. *Journal of Human-Robot Interaction* 6, 1 (2017), 25–63.
- [2] Tony Belpaeme, James Kennedy, Aditi Ramachandran, Brian Scassellati, and Fumihide Tanaka. 2018. Social robots for education: A review. *Science Robotics* 3, 21 (2018), eaat5954. <https://doi.org/10.1126/scirobotics.aat5954>
- [3] Roger Bemelmans, Gert Jan Gelderblom, Pieter Jonker, and Luc de Witte. 2012. Socially Assistive Robots in Elderly Care: A Systematic Review into Effects and Effectiveness. *Journal of the American Medical Directors Association* 13, 2 (2012), 114–120.e1. <https://doi.org/10.1016/j.jamda.2010.10.002>
- [4] James M. Berzuk and James E. Young. 2023. Clarifying Social Robot Expectation Discrepancy: Developing a Framework for Understanding How Users Form Expectations of Social Robots. In *Companion of the 2023 ACM/IEEE International Conference on Human-Robot Interaction* (Stockholm, Sweden) (HRI '23). Association for Computing Machinery, New York, NY, USA, 231–233. <https://doi.org/10.1145/3568294.3580078>
- [5] Hugh Beyer and Karen Holtzblatt. 1997. *Contextual Design: Defining Customer-Centered Systems*. Morgan Kaufmann Publishers Inc., San Francisco, CA, USA.
- [6] Louis F. Bilancia. 2014. Safe product design, forensic engineering, and Asimov's Laws of Robotics. In *2014 IEEE Symposium on Product Compliance Engineering (ISPC)*, 17–24. <https://doi.org/10.1109/ISPC.2014.6841995>
- [7] Nicholas Callander, Andrés A. Ramírez-Duque, and Mary Ellen Foster. 2024. Navigating the Human-Robot Interaction Landscape. Practical Guidelines for Privacy-Conscious Social Robots. In *Companion of the 2024 ACM/IEEE International Conference on Human-Robot Interaction* (Boulder, CO, USA) (HRI '24). Association for Computing Machinery, New York, NY, USA, 283–287. <https://doi.org/10.1145/3610978.3640637>
- [8] Laurens De Gauquier, Malaika Brengman, Kim Willems, Hoang-Long Cao, and Bram Vanderborght. 2021. In or out? A field observational study on the placement of entertaining robots in retailing. *International Journal of Retail & Distribution Management* 49, 7 (2021), 846–874. <https://doi.org/10.1108/IJRD-10-2020-0413>
- [9] Daniella DiPaola, Anastasia K. Ostrowski, Rylie Spiegel, Kate Darling, and Cynthia Breazeal. 2022. Children's Perspectives of Advertising with Social Robots: A Policy Investigation. In *2022 17th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, 570–576. <https://doi.org/10.1109/HRI53351.2022.9889572>
- [10] Francesco Del Duchetto, Paul Baxter, and Marc Hanheide. 2020. Are You Still With Me? Continuous Engagement Assessment From a Robot's Point of View. *Frontiers in Robotics and AI* 7 (2020), 116. <https://doi.org/10.3389/frobt.2020.00116>
- [11] Natalie Friedman, Awsaf Ahmed, and Wendy Ju. 2024. Skins, Clothes, Costumes, Bling: How and Why Experts Dress their Robots. In *Proceedings of the Halfway to the Future Symposium* (Santa Cruz, CA, USA) (HtF '24). Association for Computing Machinery, New York, NY, USA, Article 34, 10 pages. <https://doi.org/10.1145/3686169.3686187>
- [12] Akira Hayamizu, Michita Imai, Keisuke Nakamura, and Kazuhiro Nakadai. 2014. Volume adaptation and visualization by modeling the volume level in noisy environments for telepresence system. In *Proceedings of the Second International Conference on Human-Agent Interaction* (Tsukuba, Japan) (HAI '14). Association for Computing Machinery, New York, NY, USA, 67–74. <https://doi.org/10.1145/2658861.2658875>
- [13] Takuya Iwamoto, Jun Baba, Kotaro Nishi, Taishi Unokuchi, Daisuke Endo, Junya Nakanishi, Yuichiro Yoshikawa, and Hiroshi Ishiguro. 2021. The Effectiveness of Self-Recommendation Agents in Advancing Purchase Behavior Steps in Retail Marketing. In *Proceedings of the 9th International Conference on Human-Agent Interaction* (Virtual Event, Japan) (HAI '21). Association for Computing Machinery, New York, NY, USA, 209–217. <https://doi.org/10.1145/3472307.3484183>
- [14] Koji Kamei, Kazuhiko Shinozawa, Tetsushi Ikeda, Akira Utsumi, Takahiro Miyashita, and Norihiro Hagita. 2010. Recommendation from robots in a real-world retail shop. In *International Conference on Multimodal Interfaces and the Workshop on Machine Learning for Multimodal Interaction*, 19: 1–8. <https://doi.org/10.1145/1891903.1891929>
- [15] Takayuki Kanda, Masayuki Kamasima, Michita Imai, Tetsuo Ono, Daisuke Sakamoto, Hiroshi Ishiguro, and Yuichiro Anzai. 2007. A humanoid robot that pretends to listen to route guidance from a human. *Auton. Robots* 22, 1 (Jan. 2007), 87–100. <https://doi.org/10.1007/s10514-006-9007-6>
- [16] Foutse Khomh, Massimiliano Di Penta, Yann-Gaël Guéhenec, and Giuliano Antoniol. 2012. An exploratory study of the impact of antipatterns on class change- and fault-proneness. *Empirical Software Engineering* 17 (2012), 243–275. <https://doi.org/10.1007/s10664-011-9171-y>
- [17] Laura Kunold and Linda Onnasch. 2022. A Framework to Study and Design Communication with Social Robots. *Robotics* 11, 6 (2022). <https://doi.org/10.3390/robotics11060129>
- [18] Ali Meghdari, Minoo Alemi, Ali Ghorbandaei Pour, and Alireza Taheri. 2016. Spontaneous Human-Robot Emotional Interaction Through Facial Expressions. In *Social Robotics*, Arvin Agah, John-John Cabibihan, Ayanna M. Howard, Miguel A. Salichs, and Hongsheng He (Eds.). Springer International Publishing, Cham, 351–361.
- [19] Nikhil Muralidhar, Sathappah Muthiah, Patrick Butler, Manish Jain, Yu Yu, Katy Burne, Weipeng Li, David Jones, Prakash Arunachalam, Hays 'Skip' McCormick, and Naren Ramakrishnan. 2021. Using AntiPatterns to avoid ML Ops Mistakes. *arXiv:2107.00079* (2021). <https://doi.org/10.48550/arXiv.2107.00079>
- [20] Claudia Di Napoli and Silvia Rossi. 2019. A Layered Architecture for Socially Assistive Robotics as a Service. In *2019 IEEE International Conference on Systems, Man and Cybernetics (SMC)*, 352–357. <https://doi.org/10.1109/SMC.2019.8914532>
- [21] Marketta Niemelä, Päivi Heikkilä, and Hanna Lammi. 2017. A Social Service Robot in a Shopping Mall: Expectations of the Management, Retailers and Consumers. In *ACM/IEEE International Conference on Human-Robot Interaction*, 227–228. <https://doi.org/10.1145/3029798.3038301>
- [22] Donald A. Norman. 2013. *The Design of Everyday Things* (revised and expanded edition ed.). Basic Books, New York.
- [23] Yuki Okafuji, Jun Baba, Junya Nakanishi, Joichiro Amada, Yuichiro Yoshikawa, and Hiroshi Ishiguro. 2021. Persuasion Strategies for Social Robot to Keep Humans Accepting Daily Different Recommendations. In *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 1950–1957. <https://doi.org/10.1109/IROS51168.2021.9636772>
- [24] Yuki Okafuji, Takumi Ishikawa, Kohei Matsumura, Jun Baba, and Junya Nakanishi. 2024. Pseudo-eating behavior of service robot to improve the trustworthiness of product recommendations. *Advanced Robotics* 38, 5 (2024), 343–356. <https://doi.org/10.1080/01691864.2024.2321191>
- [25] Yuki Okafuji, Yuya Mitsui, Kohei Matsumura, Jun Baba, and Junya Nakanishi. 2023. Changes in Embarrassment Through Repeated Interactions with Robots in Public Spaces. In *2023 32nd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*, 2253–2258. <https://doi.org/10.1109/RO-MAN57019.2023.10309409>
- [26] Yuki Okafuji, Sichao Song, Jun Baba, Yuichiro Yoshikawa, and Hiroshi Ishiguro. 2023. Influence of collaborative customer service by service robots and clerks in bakery stores. *Frontiers in Robotics and AI* 10 (2023). <https://doi.org/10.3389/frobt.2023.1125308>
- [27] Eva Blessing Onyeulo and Vaibhav Gandhi. 2020. What Makes a Social Robot Good at Interacting with Humans? *Information* 11, 1 (2020). <https://doi.org/10.3390/info11010043>
- [28] OpenAI. 2023. GPT-4 Technical Report. <https://openai.com/research/gpt-4> Accessed: 2024-12-30.
- [29] Chaolong Qin, Aiguo Song, Linhu Wei, and Yu Zhao. 2023. A multimodal domestic service robot interaction system for people with declined abilities to express themselves. *Intell. Serv. Robot.* 16, 3 (June 2023), 373–392. <https://doi.org/10.1007/s11370-023-00466-6>
- [30] Julia Rosén, Jessica Lindblom, Erik Billing, and Maurice Lamb. 2021. Ethical Challenges in the Human-Robot Interaction Field. *arXiv:2104.09306 [cs.RO]* <https://arxiv.org/abs/2104.09306>
- [31] Maha Salem, Katharina Rohlfing, Stefan Kopp, and Frank Joublin. 2011. A friendly gesture: Investigating the effect of multimodal robot behavior in human-robot interaction. In *2011 RO-MAN*, 247–252. <https://doi.org/10.1109/ROMAN.2011.6005285>
- [32] T.V. Sandanger, A. Sivachandran, R.S. Eriksen, A. Baselizadeh, M.T.H. van Otterdijk, D. Kwak, and J. Torresen. 2024. Users' perception and dynamic motion velocity matching in human-robot approach scenarios. In *2024 IEEE International Conference on Advanced Robotics and Its Social Impacts (ARSO)*, 67–73. <https://doi.org/10.1109/ARSO60199.2024.10557976>
- [33] Johanna Seifert, Orsolya Friedrich, and Sebastian Schleidgen. 2022. Imitating the Human. New Human–Machine Interactions in Social Robots. *Nanoethics* 16 (2022), 181–192. <https://doi.org/10.1007/s11569-022-00418-x>
- [34] Amanda Sharkey and Noel Sharkey. 2021. We need to talk about deception in social robotics! *Ethics and Inf. Technol.* 23, 3 (2021), 309–316. <https://doi.org/10.1007/s10676-020-09573-9>
- [35] Masahiro Shiomi. 2024. A systematic survey of multiple social robots as a passive- and interactive-social medium. *Advanced Robotics* 38, 7 (2024), 440–454. <https://doi.org/10.1080/01691864.2023.2297900>
- [36] Masahiro Shiomi and Norihiro Hagita. 2016. Do Synchronized Multiple Robots Exert Peer Pressure?. In *Proceedings of the Fourth International Conference on Human Agent Interaction* (Biopolis, Singapore) (HAI '16). Association for Computing Machinery, New York, NY, USA, 27–33. <https://doi.org/10.1145/2974804.2974808>
- [37] Sichao Song, Jun Baba, Junya Nakanishi, Yuichiro Yoshikawa, and Hiroshi Ishiguro. 2021. Teleoperated Robot Sells Toothbrush in a Shopping Mall: A Field Study. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*, 221: 1–6. <https://doi.org/10.1145/3411763.3451754>
- [38] Seyyed Ehsan Salamaty Taba, Foutse Khomh, Ying Zou, Ahmed E. Hassan, and Meiyappan Nagappan. 2013. Predicting Bugs Using Antipatterns. In *2013 IEEE International Conference on Software Maintenance*, 270–279. <https://doi.org/10.1109/ICSM.2013.38>
- [39] Xiang Zhi Tan, Michal Luria, Aaron Steinfeld, and Jodi Forlizzi. 2021. Charting Sequential Person Transfers Between Devices, Agents, and Robots. In *Proceedings of the 2021 ACM/IEEE International Conference on Human-Robot Interaction*, 43–52. <https://doi.org/10.1145/3434073.3444654>

- [40] Karen Tatarián, Marine Chamoux, Amit Kumar Pandey, and Mohamed Chetouani. 2021. Robot Gaze Behavior and Proxemics to Coordinate Conversational Roles in Group Interactions. In *2021 30th IEEE International Conference on Robot & Human Interactive Communication (RO-MAN)*. 1297–1304. <https://doi.org/10.1109/RO-MAN50785.2021.9515550>
- [41] Shensheng Wang, Scott O. Lilienfeld, and Philippe Rochat. 2015. The Uncanny Valley: Existence and Explanations. *Review of General Psychology* 19, 4 (2015), 393–407. <https://doi.org/10.1037/gpr0000056>
- [42] Miki Watanabe, Kohei Ogawa, and Hiroshi Ishiguro. 2015. Can Androids Be Salespeople in the Real World?. In *Annual ACM Conference Extended Abstracts on Human Factors in Computing Systems*. 5: 251–262. <https://doi.org/10.1145/2702613.2702967>

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